

EXAMINATION PAPER — MODEL SET 1

Program:	Bachelor of Information Technology (Hons)	Module Code:	BIT 123
Module Title:	System Analysis and Design	Time Allowed:	3 Hours
Year / Semester:	Year 2 / Semester 4	Full Marks:	100 Marks

Instructions to Candidates:

- This question paper consists of **THREE (3)** sections: Section A (VSAQ), Section B (SAQ), and Section C (LAQ).
- Answer **ALL** questions in Section A. Answer **FIVE (5)** questions out of Six in Section B.
- Answer **THREE (3)** questions in Section C, where **Question No. 1 (Case Study) is COMPULSORY**.
- Candidates are required to give their answers in their own words as far as practicable.

SECTION A: VERY SHORT ANSWER QUESTIONS

[10 × 2 = 20 Marks]

Attempt ALL questions. Each question carries 2 marks.

1. Define System Analysis and mention any one primary objective of performing analysis in SDLC. Unit 1 [2]
2. State any two major characteristics of an Information System. Unit 1 [2]
3. Define Data Flow Diagram (DFD) and state its primary role in structured analysis. Unit 2 [2]
4. Mention any two standard graphical symbols used in an Entity Relationship Diagram (ERD). Unit 2 [2]
5. What is a Data Dictionary? State two entries commonly cataloged within it. Unit 3 [2]
6. Define Decision Table and identify its four essential quadrant components. Unit 4 [2]
7. State any two distinct types of feasibility studies conducted during system planning. Unit 6 [2]
8. Define Logical Design and distinguish it briefly from Physical Design. Unit 7 [2]
9. Define coupling and cohesion in the context of modular system design. Unit 7 [2]
10. Define system testing and state any two types of software maintenance. Unit 8 [2]

SECTION B: SHORT ANSWER QUESTIONS

[5 × 7 = 35 Marks]

Attempt any FIVE (5) questions out of Six. Each question carries 7 marks.

1. Explain the role of System Analysis and discuss the core technical, analytical, and interpersonal responsibilities of a System Analyst. Unit 1 & 2 [3 + 4 = 7]
2. Explain the concept of a Context Diagram (Level 0 DFD). Discuss the significance and rules of decomposing it into a Level 1 Data Flow Diagram. Unit 2 [3 + 4 = 7]
3. Discuss the purpose of Data and Process Modeling tools. Explain the role and advantages of Computer-Aided Software Engineering (CASE) tools in system development. Unit 3 [3 + 4 = 7]
4. Explain Decision Trees and Structured English. Compare their suitability with Decision Tables when modeling complex business logic. Unit 4 [3 + 4 = 7]
5. Discuss the Feasibility Study process in depth. Explain the significance of Cost/Benefit Analysis (distinguishing direct vs. indirect costs and tangible vs. intangible benefits). Unit 6 [4 + 3 = 7]
6. Explain Structured System Design. Discuss modular design, functional strength, structure charts, and techniques for minimizing module coupling. Unit 7 [3 + 4 = 7]

SECTION C: LONG ANSWER QUESTIONS

[3 × 15 = 45 Marks]

Attempt THREE (3) questions. Question No. 17 (Case Study) is COMPULSORY.

17. COMPULSORY CASE STUDY

CASE SCENARIO: ACADEMIC INSTITUTION AUTOMATION

An educational institution plans to replace its legacy manual record management system with a comprehensive, computerized **Student Information Management System (SIMS)**. Currently, student admissions, course enrollment, fee collection, grade processing, and library record transactions are managed manually on paper ledgers. This legacy operation results in severe data duplication, delayed result publication, inconsistent fee reconciliation, and frequent record misplacement. Management has appointed a Lead System Analyst to investigate the existing setup and deliver a complete architectural blueprint for the new software solution.

Based on the given scenario, answer the following questions:

- a. Identify and categorize the major stakeholders of the proposed system, detailing their primary interactions. [3]
 - b. Recommend suitable fact-finding techniques (interviews, questionnaires, observation, document sampling) to gather operational requirements. [4]
 - c. Construct a **Context Diagram (Level 0 DFD)** and a **Level 1 DFD** for the core admission and fee processing modules. [4]
 - d. Design an **Entity Relationship (ER) Diagram** displaying entities, appropriate attributes, primary keys, and relationships. [4]
8. Critically analyze the Systems Development Life Cycle (SDLC). Compare and contrast the **Waterfall, Prototyping, Spiral, and Incremental** development models in terms of risk management, cost, and requirement flexibility. Recommend and justify the most suitable model for developing an enterprise online banking portal. Unit 1 [5 + 7 + 3 = 15]
9. Discuss Object-Oriented Analysis and Design (OOAD) using UML. Explain the structural purpose, components, and behavioral notation of **Use Case Diagrams, Class Diagrams, Sequence Diagrams, and State Transition Diagrams** with appropriate diagrams. Unit 5 [3 + 4 + 4 + 4 = 15]
10. Explain the essential activities executed during the System Implementation phase. Detail system installation strategies (Direct, Parallel, Pilot, Phased), Software Quality Assurance mechanisms (formal technical reviews, walkthroughs, inspections), and the four types of software maintenance. Unit 8 [4 + 6 + 5 = 15]

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EXAMINATION PAPER — MODEL SET 2

Program:	Bachelor of Information Technology (Hons)	Module Code:	BIT 123
Module Title:	System Analysis and Design	Time Allowed:	3 Hours
Year / Semester:	Year 2 / Semester 4	Full Marks:	100 Marks

Instructions to Candidates:

- This question paper consists of **THREE (3)** sections: Section A (VSAQ), Section B (SAQ), and Section C (LAQ).
- Answer **ALL** questions in Section A. Answer **FIVE (5)** questions out of Six in Section B.
- Answer **THREE (3)** questions in Section C, where **Question No. 1 (Case Study) is COMPULSORY**.

SECTION A: VERY SHORT ANSWER QUESTIONS

[10 × 2 = 20 Marks]

Attempt ALL questions. Each question carries 2 marks.

- Differentiate between a Transaction Processing System (TPS) and a Management Information System (MIS).
Unit 1 [2]
- List any four key stakeholders involved in enterprise software development. Unit 1 [2]
- What is an external entity in a Data Flow Diagram? How is it symbolized? Unit 2 [2]
- Define cardinality and modality in Entity Relationship modeling. Unit 2 [2]
- Explain the concept of Data Store repository entries in a Data Dictionary. Unit 3 [2]
- When is Structured English preferred over a Decision Table for logic modeling? Unit 4 [2]
- What is a Use Case Actor? Give one example. Unit 5 [2]
- Define Payback Period in economic feasibility assessment. Unit 6 [2]
- State two primary objectives of effective user interface and input form design. Unit 7 [2]
- Distinguish between Alpha Testing and Beta Testing. Unit 8 [2]

SECTION B: SHORT ANSWER QUESTIONS

[5 × 7 = 35 Marks]

Attempt any FIVE (5) questions out of Six. Each question carries 7 marks.

- Examine the evolving role of a System Analyst in modern organizations. Discuss how their communication and analytical skills bridge business requirements and technical implementation. Unit 1 & 2 [3 + 4 = 7]
- Explain the rules and conventions governing DFD construction. What constitutes "Black Hole," "Miracle," and "Gray Hole" errors in process modeling? Unit 2 [3 + 4 = 7]
- Discuss the concept of Data Dictionaries. Explain how data structures, data elements, and data stores are formally documented. Unit 3 [3 + 4 = 7]
- Construct a Decision Table for a retail discount policy: "Give 15% discount if purchase > \$500 and member; 10% if purchase > \$500 but non-member; 5% if member with purchase ≤ \$500; else no discount." Unit 4 [7]
- Explain requirement gathering techniques including Joint Application Development (JAD) and rapid questionnaires. Discuss their comparative merits. Unit 6 [3 + 4 = 7]
- Discuss the differences between Logical Design and Physical Design in database modeling and input/output structure architecture. Unit 7 [3 + 4 = 7]

SECTION C: LONG ANSWER QUESTIONS

[3 × 15 = 45 Marks]

Attempt THREE (3) questions. Question No. 17 (Case Study) is COMPULSORY.

17. COMPULSORY CASE STUDY

CASE SCENARIO: HOSPITAL INFORMATION & APPOINTMENT SYSTEM

A multi-specialty healthcare facility is transitioning from paper-based scheduling and manual patient records to a centralized **Hospital Information and Appointment System (HIAS)**. The system must allow patients to register, search doctor schedules, and book appointments online. Doctors must be able to view their daily schedule, update electronic medical records (EMRs), and issue electronic prescriptions. The diagnostic laboratory needs to upload lab investigation reports directly to the patient's record, while the accounts division must manage automated billing and insurance claims.

Based on the given scenario, answer the following questions:

- a. Identify all primary external entities and compile functional and non-functional requirements for the hospital platform. [4]
 - b. Draw a **Level 0 Context Diagram** demonstrating data flows between external entities and HIAS. [4]
 - c. Construct a **Level 1 DFD** focusing on the *Appointment Scheduling* and *Electronic Prescription* sub-processes. [4]
 - d. Perform a structured feasibility assessment evaluating Technical, Operational, Economic, and Schedule viability. [3]
8. Explain the concept of software architecture modularity. Critically discuss the spectrum of **Cohesion** (from Coincidental to Functional) and **Coupling** (from Data to Content). Explain how high cohesion and loose coupling contribute to maintainability and fault isolation. Unit 7 [5 + 5 + 5 = 15]
 9. Discuss object modeling with Unified Modeling Language (UML). Explain the distinction between structural and behavioral models. Draw and explain a comprehensive **Class Diagram** and an accompanying **Sequence Diagram** for an Online Shopping Cart checkout process. Unit 5 [5 + 5 + 5 = 15]
 10. Define System Testing. Differentiate systematically between **Unit Testing, Integration Testing, System Testing, and Acceptance Testing (UAT)**. Explain the role of regression testing and formal technical reviews (FTR) in maintaining software quality assurance. Unit 8 [8 + 4 + 3 = 15]

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EXAMINATION PAPER — MODEL SET 3

Program:	Bachelor of Information Technology (Hons)	Module Code:	BIT 123
Module Title:	System Analysis and Design	Time Allowed:	3 Hours
Year / Semester:	Year 2 / Semester 4	Full Marks:	100 Marks

Instructions to Candidates:

- This question paper consists of **THREE (3)** sections: Section A (VSAQ), Section B (SAQ), and Section C (LAQ).
- Answer **ALL** questions in Section A. Answer **FIVE (5)** questions out of Six in Section B.
- Answer **THREE (3)** questions in Section C, where **Question No. 1 (Case Study) is COMPULSORY**.

SECTION A: VERY SHORT ANSWER QUESTIONS

[10 × 2 = 20 Marks]

Attempt ALL questions. Each question carries 2 marks.

- State two main limitations of the traditional Waterfall Model of SDLC. Unit 1 [2]
- What is meant by an Information System's boundary and environment? Unit 1 [2]
- Differentiate between functional requirements and non-functional requirements. Unit 6 [2]
- What is a recursive relationship in an Entity Relationship Diagram? Unit 2 [2]
- Define Upper-CASE vs. Lower-CASE tools in system modeling. Unit 3 [2]
- Mention the purpose of an Extended Entry Decision Table. Unit 4 [2]
- State the purpose of a State Machine (State Transition) Diagram in UML. Unit 5 [2]
- What is Return on Investment (ROI) and how is it expressed mathematically? Unit 6 [2]
- What is a Structure Chart and what do arrows with open/solid circles indicate? Unit 7 [2]
- Define preventive software maintenance. Unit 8 [2]

SECTION B: SHORT ANSWER QUESTIONS

[5 × 7 = 35 Marks]

Attempt any FIVE (5) questions out of Six. Each question carries 7 marks.

- Explain the major phases of the Systems Development Life Cycle (SDLC) and discuss the primary deliverables produced at the end of each phase. Unit 1 [7]
- Explain the procedure for drawing an Entity Relationship Diagram (ERD). Describe entity types, weak entities, attributes, and key constraints with examples. Unit 2 [3 + 4 = 7]
- Describe the structure and management of a Project Repository. Explain how Data Dictionaries enforce consistency across analysis and design models. Unit 3 [3 + 4 = 7]
- Differentiate between Decision Tables and Decision Trees. Draw a Decision Tree to represent loan eligibility:
 Salary ≥ \$50k grants approval; Salary < \$50k with collateral grants approval; otherwise reject. Unit 4 [3 + 4 = 7]
- Explain the concepts of Tangible vs. Intangible Costs and Benefits. Provide two distinct real-world examples for each category in corporate systems. Unit 6 [3.5 + 3.5 = 7]
- Discuss Input and Form Design techniques. What design principles should be followed to ensure data integrity, user ergonomics, and minimal data entry error? Unit 7 [3 + 4 = 7]

SECTION C: LONG ANSWER QUESTIONS

[3 × 15 = 45 Marks]

Attempt THREE (3) questions. Question No. 17 (Case Study) is COMPULSORY.

17. COMPULSORY CASE STUDY

CASE SCENARIO: E-COMMERCE RETAIL & INVENTORY MANAGEMENT

A retail enterprise is upgrading its nationwide supply chain with an integrated **E-Commerce & Warehouse Inventory Platform**. The system must support customer account management, dynamic catalog browsing, secure checkout via

third-party payment gateways, and real-time inventory decrementing. When stock levels reach a critical reorder point, the inventory subsystem must automatically draft purchase orders for suppliers. The logistics department requires dispatch manifests, shipping tracking integration, and return RMA processing.

Based on the given scenario, answer the following questions:

- a. Identify all external entities and develop a comprehensive **Context Diagram (Level 0 DFD)** for the system. [4]
 - b. Construct a **Level 1 DFD** detailing: Order Processing, Payment Verification, and Inventory Replenishment. [4]
 - c. Design a detailed **ER Diagram** showing entities (Customer, Order, Product, Supplier, Shipment), their attributes, and relationships with cardinalities. [5]
 - d. Explain how data dictionary entries would be defined for the data store D1: Product_Inventory. [2]
8. Discuss Project Scheduling and Feasibility Analysis in System Planning. Explain the use of **Gantt Charts** and **PERT/CPM** networks in monitoring project milestones. Given a hypothetical system development project, explain how the Critical Path is determined and managed. Unit 6 [5 + 5 + 5 = 15]
9. Explain the principles of Database Design in system engineering. Discuss the step-by-step transition from an unnormalized data set through **First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF)**. Why is normalization indispensable for enterprise systems? Unit 7 [3 + 9 + 3 = 15]
10. Explain the System Conversion and Installation phase. Compare **Direct Cutover, Parallel Operation, Pilot Operation, and Phased Rollout** in terms of operational risk, resource overhead, and failure recovery. Discuss post-implementation review procedures. Unit 8 [10 + 5 = 15]

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EXAMINATION PAPER — MODEL SET 4

Program:	Bachelor of Information Technology (Hons)	Module Code:	BIT 123
Module Title:	System Analysis and Design	Time Allowed:	3 Hours
Year / Semester:	Year 2 / Semester 4	Full Marks:	100 Marks

Instructions to Candidates:

- This question paper consists of **THREE (3)** sections: Section A (VSAQ), Section B (SAQ), and Section C (LAQ).
- Answer **ALL** questions in Section A. Answer **FIVE (5)** questions out of Six in Section B.
- Answer **THREE (3)** questions in Section C, where **Question No. 1 (Case Study) is COMPULSORY**.

SECTION A: VERY SHORT ANSWER QUESTIONS

[10 × 2 = 20 Marks]

Attempt ALL questions. Each question carries 2 marks.

1. What is an Executive Support System (ESS) and who are its primary organizational users? **Unit 1** [2]
2. Explain the concept of "Data Flow Splitting" and "Data Flow Merging" in DFDs. **Unit 2** [2]
3. What is a composite attribute in conceptual data modeling? Give an example. **Unit 2** [2]
4. What is metadata and how is it documented within a data dictionary? **Unit 3** [2]
5. State the meaning of "Indifference Condition" in a Decision Table. **Unit 4** [2]
6. Define generalization and specialization in UML Class Diagrams. **Unit 5** [2]
7. What is Operational Feasibility and why is user acceptance central to it? **Unit 6** [2]
8. Define stamp coupling and control coupling. **Unit 7** [2]
9. What is a Structure Chart "Module Call"? **Unit 7** [2]
10. Distinguish between Adaptive Maintenance and Perfective Maintenance. **Unit 8** [2]

SECTION B: SHORT ANSWER QUESTIONS

[5 × 7 = 35 Marks]

Attempt any FIVE (5) questions out of Six. Each question carries 7 marks.

1. Compare Structured Analysis & Design (SAD) with Object-Oriented Analysis & Design (OOAD). Discuss their foundational modeling paradigms. **Unit 1 & 5** [3 + 4 = 7]
2. Explain the concept and significance of DFD Balancing (level-to-level child-parent data flow consistency) with an illustrative diagram. **Unit 2** [3 + 4 = 7]
3. Discuss the role of CASE tools in repository management, diagram generation, code skeleton generation, and traceability throughout the SDLC. **Unit 3** [3 + 4 = 7]
4. Explain the constructs used in Structured English (sequence, selection, iteration). Transform a complex multi-condition business rule into Structured English. **Unit 4** [3 + 4 = 7]
5. Discuss the steps involved in performing a formal Economic Feasibility analysis. Explain Net Present Value (NPV) and Break-Even Analysis. **Unit 6** [3 + 4 = 7]
6. Explain Software Quality Assurance (SQA). Compare formal Code Inspections, Structured Walkthroughs, and Technical Reviews in defect identification. **Unit 8** [3 + 4 = 7]

SECTION C: LONG ANSWER QUESTIONS

[3 × 15 = 45 Marks]

Attempt THREE (3) questions. Question No. 17 (Case Study) is COMPULSORY.

17. COMPULSORY CASE STUDY

CASE SCENARIO: AUTOMATED PUBLIC TRANSIT & SMART TICKETING

A regional metropolitan transit corporation is designing a centralized **Smart Transit & Automated Ticketing System (STATS)**. Commuters can purchase RFID smart cards, top up balances via mobile banking, and tap in/tap out at bus

terminals with automated distance-based fare deductions. Bus drivers log in to on-board GPS terminals transmitting live route coordinates to a central dispatch monitor. Fleet controllers monitor scheduling delays and issue reroute dispatches. Management requires daily passenger traffic metrics and fare revenue auditing reports.

Based on the given scenario, answer the following questions:

- a. Identify all system stakeholders and state three functional and two non-functional requirements. [4]
 - b. Construct a **Context Diagram (Level 0 DFD)** modeling interaction between Commuters, Drivers, Dispatchers, and Banking Gateways. [4]
 - c. Draw a comprehensive **Use Case Diagram** illustrating primary use cases, actor associations, and «include» / «extend» relationships. [4]
 - d. Propose a deployment installation strategy for migrating from paper tickets to RFID smart gates with minimum passenger disruption. [3]
8. Critically evaluate the **Agile Development Methodology** (Scrum/XP) versus the **Spiral Model**. Contrast their sprint/iterative delivery, client involvement, risk handling, and artifact documentation. Discuss when an organization should choose Agile over structured SDLC. Unit 1 [5 + 5 + 5 = 15]
9. Discuss Object Modeling with UML in detail. Explain **Sequence Diagrams** and **Collaboration (Communication) Diagrams**. Draw a Sequence Diagram showing object interactions (Lifelines, Messages, Activation Bars) for a User Authentication and Password Reset workflow. Unit 5 [5 + 5 + 5 = 15]
10. Explain the principles of User Interface (UI) and Output Design. Discuss human factors in UI engineering, screen layout structuring, data validation controls, and error alerting strategies. Explain the importance of output forms and report generation for management decision-making. Unit 7 [7 + 4 + 4 = 15]

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